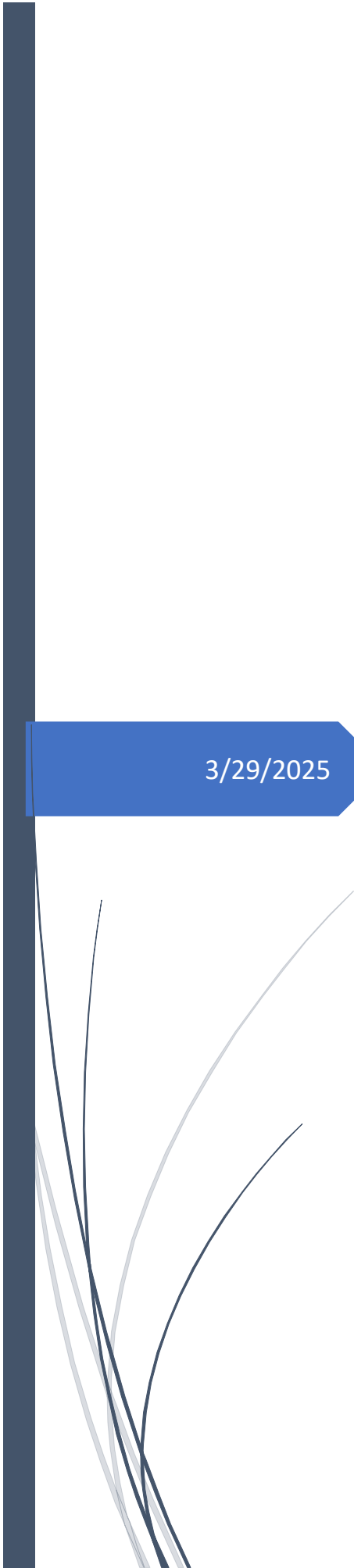




HEALTHCARE ACCESSIBILITY REPORT

PROJECT 3 SUBMISSION

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OUTLINES

- I. Introduction
- II. Key Disparities in Healthcare Access (Urban vs. Rural)
- III. Efficiency of Healthcare Funding and Resource Allocation
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- V. Strategic Recommendations for Policymakers
- VI. Data Visualization for Stakeholder Communication

1. INTRODUCTION:

This case study aims to analyze healthcare facility data, identify disparities in access to healthcare between rural and urban regions, and evaluate the efficiency of healthcare funding. The objective is to propose data-driven recommendations that governments and stakeholders can implement to enhance healthcare service delivery across Africa.

A separate column was created using conditional column to identify urban and rural area by giving the condition of “if annual patient visit is less than or equal to 8000 then rural, if greater than or equal to 8001 then urban.

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name
Rural/Urban

	Column Name	Operator	Value		Output
If	Annual Patient Vis...	is less than or equ...	8000	Then	Rural
Else If	Annual Patient Vis...	is greater than or...	8001	Then	Urban

An overview of the conditional column

2. KEY DISPARITIES IN HEALTHCARE ACCESS (URBAN VS. RURAL)

- **Healthcare workforce imbalance:**
 - Urban areas have 13,542 doctors and 35,048 nurses.
 - Rural areas have only 7,559 doctors and 19,607 nurses—less than half the number of urban healthcare professionals.
- **Infrastructure challenges:**
 - Internet and electricity availability are significantly lower in rural areas, limiting access to telemedicine and digital health records.
 - The average distance to healthcare facilities is much higher in rural regions, making access more difficult and this has to be looked into because we know that means of transportation in some remote villages isn't as fast as we have in urban area

- **Patient visits and hospital facilities:**

- Urban areas handle 17.95 million patient visits annually, compared to 3.23 million in rural areas and this is as a result of rural_urban migration and low believe system in hospitals in rural areas
- Despite fewer visits, rural areas experience longer wait times

3. EFFICIENCY OF HEALTHCARE FUNDING AND RESOURCE ALLOCATION

- **Funding disparities:**

- Urban areas receive \$135.7 million, while rural areas receive \$77.4 million.
- This shows a 64% vs. 36% split, similar to the population distribution.

4. EMERGENCY RESPONSE AND PATIENT SATISFACTION

- **Response times:**

- Rural emergency response times are much longer compared to urban areas.

- **Patient satisfaction levels:**

- Urban patient satisfaction rates are noticeably higher, as seen in faster response times and better facility conditions.
- Rural patient dissatisfaction could result from long travel distances, poor internet, and lower service availability.

5. STRATEGIC RECOMMENDATIONS FOR POLICYMAKERS

1. Increase rural healthcare funding:

- Adjust allocations to so that rural areas can equally get a reasonable funding so as to improve their facilities and services so as to also reduce the movement of people from rural areas to urban area to seek good health care
- Provide incentives for healthcare workers to serve in rural regions.

2. Improve rural emergency response systems:

- Deploy ambulance networks and integrate community-led first-aid programs.
- Enhance telemedicine services to connect remote areas with urban specialists.
- 3. Expand rural healthcare facilities:**
 - Fund more rural clinics and health centres at strategic locations, reducing the burden on urban hospitals.
- 4. Enhance electricity and internet access:**
 - Provide energy solutions like the use of solar power for rural hospitals.
 - Invest in internet infrastructure to enable telemedicine and digital patient records.
- 5. Monitor patient feedback and satisfaction:**
 - Develop a nationwide patient satisfaction tracking system to guide improvements.
 - Use data-driven assessments to prioritize underperforming regions.

6. DATA VISUALIZATION FOR STAKEHOLDER COMMUNICATION

- **Disparity in workforce allocation:** Graphs highlighting doctor/nurse distribution per region.
- **Funding allocation breakdown:** Visuals showing urban vs. rural funding gaps by facility type.
- **Emergency response impact:** Charts depicting response time vs. patient satisfaction.
- **Internet and electricity availability:** Demonstrating the infrastructure divide affecting rural healthcare.

Here is the link the dashboard for healthcare access

https://drive.google.com/file/d/1MKdmv0dd5o5_r90kb4FySk6ZdljJR2jW/view?usp=sharing